6CCSAACV
Advanced Computer Vision

Revision Notes

Academic Year 2025/26 — Semester 2

Compiled on January 28, 2026

These notes are structured to help you prepare for exams.
Each week includes: learning objectives, core concepts, theory, and worked examples.

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Module Overview

About This Module

This module provides students with a comprehensive understanding of the fundamental concepts and techniques involved in **interpreting and extracting information from visual data**.

Module code: 6CCSAACV **Semester:** 2 (AY 2025/26)

Module Lecturer: Dr. Luis Carlos Garcia Peraza Herrera

Module Aims

The module aims to:

- Provide a comprehensive understanding of fundamental concepts and techniques in computer vision.
- Develop skills in interpreting and extracting information from visual data.
- Cover both traditional and state-of-the-art machine learning techniques for vision tasks.

Learning Outcomes

By the end of this module, you will be able to:

Code	Learning Outcome
ACV1	Identify and solve complex computer vision tasks, selecting and implementing the most suitable algorithms and methodologies.
ACV2	Design and develop innovative methods and algorithms for addressing various computer vision challenges, critically evaluating their strengths and limitations.
ACV3	Analyse, assess, and effectively deploy a range of traditional and state-of-the-art machine learning techniques to implement robust solutions to diverse computer vision tasks.

Recommended Reading

There is no required reading for this module. However, the following books are recommended:

1. *Computer Vision: Algorithms and Applications* (2nd ed). Richard Szeliski. Springer, 2022.
2. *Seeing: The Computational Approach to Biological Vision* (2nd ed). John P. Frisby and James V. Stone. MIT Press, 2010.
3. *Computer Vision: Models, Learning, and Inference.* Simon J. D. Prince. Cambridge University Press, 2012.
4. *Deep Learning: Foundations and Concepts.* Christopher Bishop. Springer, 2023.

Assessment

[TODO] Add assessment details when provided (coursework + exam breakdown).

Weekly Topics

[TODO] Add weekly topic overview once lecture slides are provided.

1 Week 1: Introduction

1.1 Topic Summary

Learning Objectives

By the end of this week, you will be able to:

- Describe the goals and challenges of the Computer Vision field.
- Recognise real-world scenarios where Computer Vision is used.
- Identify and list various Computer Vision applications.

Big Picture. This week introduces the field of Computer Vision: what it is, why it matters, why it's hard, and how we approach solving vision problems. Vision is our primary sense for experiencing the world, yet teaching machines to "see" remains one of the great challenges in AI. The key insight is that vision is an ***ill-posed inverse problem***—many possible 3D scenes can produce the same 2D image. Solving this requires ***priors*** (constraints/expectations) that make some interpretations more likely than others.

What Examiners Like to Ask:

- Define computer vision and distinguish it from related fields (image processing, computer graphics).
- Explain why vision is an "ill-posed inverse problem."
- Give examples of priors and how they constrain interpretations.
- List and categorise CV applications by type of information extracted.
- Explain why deep learning is not always the best solution.

1.2 Core Concepts

Roadmap:

1. What is Computer Vision? (definitions, related fields)
2. Why study vision? (importance, applications)
3. Why is vision difficult? (ill-posed, exponentially large)
4. How do we tackle the problem? (priors, engineering approaches)

1.2.1 What is Computer Vision?

Computer Vision

Computer Vision is a field of study that focuses on creating methods capable of extracting information about the world from images and videos.

Classic definitions:

- "Computing properties of the 3D world from one or more digital images" (Trucco & Verri)
- "Making useful decisions about real physical objects and scenes based on sensed images" (Stockman & Shapiro)
- "Extracting descriptions of the world from pictures or sequences of pictures" (Forsyth & Ponce)

Alternative names: machine vision, image analysis, image understanding, computational vision.

Related Disciplines and Their Transformations:

Field	Input	→	Output
Image Processing	Image	→	Image
Computer Vision	Image	→	Description
Computer Graphics	Description	→	Image

Exam Tip: Field Distinctions

A common question asks you to label arrows in a diagram showing transitions between "Real World," "Image," and "Description." Remember:

- Real World → Image: **Imaging/Photography**
- Image → Description: **Computer Vision**
- Description → Image: **Computer Graphics**
- Image → Image: **Image Processing**

Other related fields:

- **Pattern Recognition**: recognising and classifying stimuli in images and other datasets.
- **Photogrammetry**: obtaining measurements from images.
- **Biological Vision**: understanding visual perception in humans/animals (neuroscience, psychology, psychophysics).

1.2.2 Why Study Computer Vision?

Motivation:

- Vision is the main way we experience the world.
- Want machines to interact with and understand the world (robotics, AI).
- Digital images are everywhere—lots of information to extract.
- Many practical applications with real-world impact.

Applications by Information Extracted:

Application	Information Extracted
OCR / Character Recognition	Category of each letter/digit
Face Tracking (webcam)	Location of face in image
Face Recognition / Biometrics	Identity of individual
Content-Based Image Retrieval	Category of items, relationships between items
Driver Assistance	Categorisation (pedestrians, road edge) + 3D location
3D Modelling	3D locations of all image points
Object Detection	Location + class of objects
Object Segmentation	Pixel-level object boundaries
Action Recognition	Temporal classification of actions
Visual SLAM	Camera pose + 3D map of environment

Application Categories:

- **Recognition**: OCR, face recognition, iris/fingerprint biometrics, landmark recognition
- **Detection & Tracking**: face detection, people tracking, object tracking (e.g., Hawk-Eye in sports)
- **Retrieval**: content-based image retrieval, reverse image search
- **3D Reconstruction**: 3D models from images, structure from motion
- **Autonomous Systems**: driver assistance, self-driving cars, space exploration (Mars rovers)
- **Medical**: analysis of MRI, CT, ultrasound; surgical robotics
- **Entertainment**: AR/VR, TikTok filters, eye contact rectification
- **Industrial**: quality control inspection, barcode/QR scanning

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1.2.3 Why is Computer Vision Difficult?

Two Major Challenges

1. **One image → many interpretations**: the problem is ***ill-posed***.
2. **One object → many images**: the problem is ***exponentially large***.

Challenge 1: Vision is an Ill-Posed Inverse Problem

Forward vs Inverse Problems

- **Forward problem**: know causes, predict outcomes (well-posed, unique solution).
- **Inverse problem**: know outcomes, infer causes (often ill-posed, multiple solutions).

- **Imaging** (forward): 3D world → 2D image is **unique** (well-posed).
- **Vision** (inverse): 2D image → 3D world is **NOT unique** (ill-posed).

For any given image, many different 3D scenes could have generated it. Humans typically perceive only one interpretation because our brains apply ***priors*** that make some interpretations more likely.

Classic Ambiguous Figures

- **Necker Cube**: two equally valid 3D interpretations; perception flips spontaneously.
- **Rubin's Vase**: face or vase? Two interpretations, never perceived simultaneously.
- **Convex/Concave Craters**: prior expectation of light from above determines interpretation.

Challenge 2: Vision Scales Exponentially

Consider recognising an object that can appear at:

- l locations in the image
- s different scales (sizes)
- o orientations
- c colours

This single object can generate $l \times s \times o \times c$ different images. The number of possible images grows **exponentially** with the number of parameters.

Factors Affecting Appearance:

Factor	Effect
Viewpoint	Same object from different angles looks very different
Illumination	Different lighting changes appearance drastically
Non-rigid deformation	Object can bend, stretch (e.g., faces, animals)
Within-category variation	Objects in same category look different (e.g., different chairs)
Background clutter	Multiple objects in scene
Occlusion	Objects partially hidden by other objects

Exam Tip: Discrimination vs Recognition

"Objects that look very similar can be represented and recognized as different objects, whereas objects that look very different can be recognized as the same basic-level objects" (Bar, 2004). This captures the dual challenge: distinguishing similar things while grouping dissimilar things.

Why Current Systems Are Limited:

- Most CV systems work in **specific domains**: specific task + specific environment.
- Solutions are **not robust** and **do not generalise** well.
- Code that works for one task may fail for similar tasks or same task under different conditions.
- The challenge: develop **general-purpose vision systems** that match human performance.

Challenging Examples

- **Chihuahua or blueberry muffin?** Texture similarity fools classifiers.
- **Sheepdog or mop?** Shape/texture confusion.
- **How many cows?** Segmentation fails with unusual poses/occlusions.

These examples show that even state-of-the-art deep learning struggles with adversarial or unusual cases.

1.2.4 How Do We Tackle the Problem of Vision?

Prior (Constraint)

A ***prior*** is additional information that helps reduce the number of possible solutions, making some interpretations more likely than others. Priors are essential for solving the ill-posed inverse problem of vision.

Sources of Priors:

1. **Prior knowledge / familiarity**:
   - Learned familiarity with certain objects
   - Knowledge of image formation process (e.g., perspective, shadows)
2. **Prior exposure / motion / priming**:
   - Recent or preceding sensory input
   - Change blindness: disrupting temporal continuity hides changes
3. **Current context**:
   - Surrounding visual scene
   - Concurrent input from other senses

Effects of Priors on Human Perception:

Prior Type	Example Effect
Illumination	Checker shadow illusion: squares A and B appear different but are same intensity (shadow prior)
Light direction	Craters appear convex/concave depending on assumed light source (from above)
Perspective	Ames room: people appear to change size due to perspective expectations
Prior knowledge	Dalmatian dog: invisible until you know what to look for
Context	"THE CAT"—middle letter is H in first word, A in second (same shape)
Face detection	We see faces everywhere (Virgin Mary toast sold for $28,000!)

Illusions Reveal Assumptions

Illusions are not "mistakes" by the visual system. They reveal the **assumptions/priors** the visual system uses to solve the under-constrained problem of vision. These assumptions work well in typical situations because they reflect how the world usually is.

Two Engineering Approaches:

Approach	Description
Forward Engineering	Determine requirements → design system → implement and refine. "Top-down": start with computational theory, fill in details.
Reverse Engineering	Find a system that works (e.g., brain) → analyse how it works → implement using same mechanisms. "Bottom-up": start with mechanisms, build model.

This course considers **both** approaches:

- **Computational (machine) vision**: How can we get computers to see?
- **Biological (human) vision**: How do people see?

Exam Tip: Deep Learning Is Not Always the Answer

The lecture emphasises that traditional/classical CV methods sometimes outperform deep learning. Example: for endoscope content area segmentation, classical methods achieved better accuracy than DL approaches. Always consider the problem domain and data availability.

1.3 Worked Examples

Example 1: Defining Computer Vision

Q: Give a definition of "Computer Vision."

A: Computer Vision is a field of study that focuses on creating methods capable of extracting information about the world from images and videos.

Example 2: Field Transitions Diagram

Q: Label each arrow with the field that deals with that transition:

$$\text{Real World} \leftrightarrow \text{Image} \leftrightarrow \text{Description}$$

A:

- Real World → Image: **Imaging/Photography**
- Image → Real World: (Physical interaction, not a field)
- Image → Description: **Computer Vision**
- Description → Image: **Computer Graphics**
- Image → Image: **Image Processing**

Example 3: Information Extracted by Application

Q: What type of information is extracted in each application?

A:

- **OCR**: category of each letter
- **Face tracking webcam**: location of face in image
- **Face recognition/biometrics**: identity of individual
- **Content-based image retrieval**: category of items, relationships between items
- **Driver assistance**: categorisation (pedestrians, road edge) + 3D location

- **3D modelling**: 3D locations of all image points

Example 4: Ill-Posed Inverse Problem

Q: Vision is often described as an "ill-posed, inverse problem." Explain these terms and their opposites.

A:

- **Forward problem**: Know the causes, predict the outcomes (e.g., know forces, calculate acceleration).
- **Inverse problem**: Know the outcomes, infer the causes (e.g., know pixel intensities, infer 3D scene).
- **Well-posed problem**: Has one unique solution.
- **Ill-posed problem**: Has multiple solutions (or no solution).

Vision is an **inverse problem** because we know pixel intensities (outcomes) and want to infer the 3D scene (causes). It is **ill-posed** because multiple 3D scenes can produce the same 2D image.

Example 5: Priors in Vision

Q: What is a "prior" and how does it help solve the ill-posed inverse problem of vision? Give examples.

A: A prior is additional information that reduces the number of possible solutions, making some interpretations more likely. Examples include:

- **Knowledge of the world**: objects are usually solid, light comes from above
- **Image formation knowledge**: perspective geometry, how shadows work
- **Prior exposure**: recently seen objects are more easily recognised
- **Context**: surrounding scene constrains possible object identities

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1.4 Summary

Key Takeaways

1. **Computer Vision** extracts descriptions/information from images (Image → Description).
2. Vision is **difficult** because it is:
   - **Ill-posed**: one image can have many 3D interpretations
   - **Exponentially large**: one object can generate many images
3. **Priors/constraints** are essential for narrowing down interpretations.
4. Human vision uses priors extensively—illusions reveal these assumptions.
5. We tackle CV using both **forward engineering** (design systems) and **reverse engineering** (study biological vision).
6. Deep learning is powerful but **not always the best solution**—classical methods can outperform in specific domains.

Concept	**Definition**	**Implication**
Inverse problem	Infer causes from outcomes	Multiple solutions possible
Ill-posed	No unique solution	Need priors to constrain
Prior	Additional constraining info	Makes some interpretations likely
Forward engineering	Top-down design	Computational theory first
Reverse engineering	Bottom-up modelling	Study biological systems

2 Week 2: Image Formation

2.1 Topic Summary

Learning Objectives

By the end of this week, you will be able to:

- Explain the end-to-end process of digital image capture.
- Analyse the biological mechanisms of visual perception in the human eye.
- Compare the different strategies used by digital cameras and the human retina to efficiently encode spatial and colour information.

Big Picture. This week covers how images are formed—both in cameras and in the eye. We examine the ***physics*** (light, reflectance, optics), ***geometry*** (projection, camera models), ***digitisation*** (sampling, quantization), and ***biological vision*** (photoreceptors, ganglion cells). The key insight is that recovering surface properties from images is ill-posed because $L = f(E, R)$—we observe luminance but need to infer reflectance without knowing illumination.

What Examiners Like to Ask:

- Explain the relationship $L(\lambda) = f(E(\lambda), R(\lambda))$ and why colour constancy is hard.
- Derive the perspective projection equations and explain intrinsic/extrinsic parameters.
- Compare pinhole cameras vs lensed cameras (trade-offs).
- Explain demosaicing and different interpolation methods.
- Describe centre-surround receptive fields and their functional benefits.
- Compare foveal vs peripheral vision.

2.2 Core Concepts

Roadmap:

1. Physics of image formation (light, reflectance, optics)
2. Geometry of image formation (camera models, projective geometry)
3. Digital images (digitisation, representation)
4. Camera sensors (CCD, demosaicing)
5. The eye (photoreceptors, ganglion cells)

2.2.1 Physics of Image Formation

Overview of Image Formation:

Images are formed when a **sensor** registers **radiation** that has interacted with **physical objects**. Light from a source is reflected by surfaces, passes through optics, and is recorded by a sensor.

$$\text{Light Source} \rightarrow \text{Surface (Reflectance)} \rightarrow \text{Optics} \rightarrow \text{Sensor}$$

Two Sets of Parameters:

Parameter Type	What It Determines
Radiometric	Intensity/colour at each image location (illumination, surface reflectance, sensor sensitivity)
Geometric	Where scene points appear in image (camera position/orientation, optics, projection geometry)

Light and Colour:

Visible Light

Electromagnetic radiation with wavelength λ between 400–700nm. The sun's intensity peaks in this range, and human eyes evolved sensitivity to it.

- **Illuminance** $E(x, y, \lambda)$: amount of light striking a surface (varies by light source)
- **Reflectance** $R(x, y, \lambda)$: fraction of light reflected at each wavelength (***albedo***)
- **Luminance** $L(x, y, \lambda)$: amount of light reaching the sensor

Fundamental Equation of Image Formation

$$L(x, y, \lambda) = f\big(E(x, y, \lambda), R(x, y, \lambda)\big)$$

The luminance depends on both illuminance and reflectance. This is **ill-posed**: we observe L but need to recover R without knowing E.

Colour Mixing:

Type	Process	Result
Additive (light)	R + G + B lights	White (broad spectrum)
Subtractive (pigment)	R + G + B pigments	Black (absorbs all)

Why Objects Have Colour

An object looks **green** because it absorbs red and blue light, leaving more green in the reflected light. The colour we perceive depends on which wavelengths are *not* absorbed.

Colour Constancy:

Colour Constancy

The ability to perceive surface colour as constant despite changes in illumination. Humans do this well; computers struggle.

Approaches to estimate illumination E:

- Assume average reflectance across scene is known (grey world)
- Fix brightest patch to white
- Gamut falls within known range
- Known reference colour (colour chart, skin)
- Specular reflections have illumination colour

None work reliably in all cases. Human vision discounts illumination effects through learned priors.

2.2.2 Optics: Pinhole and Lens Cameras

Pinhole Camera

A camera that restricts light flow through a small hole so that only one ray from each scene point reaches the sensor. Everything is in focus regardless of depth.

Pinhole Trade-off:

Aperture	Focus	Brightness
Small pinhole	Sharp	Dim (long exposure, blur from motion)
Large pinhole	Blurred	Bright (short exposure)

A **lens** solves this by focusing rays from the same point onto a single image location, keeping the image sharp while gathering more light.

Thin Lens Properties:

- Rays through optical centre O are not refracted
- Rays parallel to optical axis refract through focal point F'
- Rays through F become parallel to optical axis

Thin Lens Equation

$$\frac{1}{f} = \frac{1}{|z|} + \frac{1}{|z'|}$$

where f = focal length, z = object distance, z' = image distance.

Key implications:

- Object at infinity $\Rightarrow$ image at F'
- Object at F or closer $\Rightarrow$ no image formed
- **Depth of focus**: range of object depths that appear acceptably sharp
- Smaller aperture $\Rightarrow$ larger depth of focus (approaches pinhole behaviour)

Exam Tip: Lens vs Pinhole

Know the trade-offs: pinhole has infinite depth of focus but needs long exposure; lens is bright but only objects at specific depths are in focus.

2.2.3 Geometry of Image Formation

Perspective (Pinhole) Camera Model

A mathematical model of geometric projection where a 3D point $P = (x, y, z)$ projects to image point $P' = (x', y', z')$. The image plane Π' is at distance f' from the pinhole along the optical axis.

Equation of Projection:

From similar triangles:

$$\frac{x'}{x} = \frac{f'}{z} \quad \Rightarrow \quad x' = \frac{f'}{z}x$$

$$\frac{y'}{y} = \frac{f'}{z} \quad \Rightarrow \quad y' = \frac{f'}{z} y$$

Since P' lies on the image plane: $z' = f'$.

In matrix form (homogeneous coordinates):

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \frac{f'}{z} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Projective Geometry:

Geometry	Describes	Preserves
Euclidean	Objects "as they are" (3D→3D)	Lengths, angles, parallelism
Projective	Objects "as they appear" (3D→2D)	None of the above

Projective distortions:

- Distant objects appear smaller
- Parallel lines converge at ***vanishing points***
- Equal lengths become unequal
- Right angles become non-right angles

Vanishing Point

The projection of a point at infinity. All parallel lines in 3D share the same vanishing point in the 2D image.

Railway Tracks Example

In the world: tracks parallel, ties equal length, perpendicular, evenly spaced.

In the image: tracks converge at vanishing point, ties get shorter with distance, not perpendicular to tracks, get closer together.

2.2.4 Intrinsic and Extrinsic Parameters

Converting Camera Coordinates to Pixels:

Intrinsic Parameters

Camera-dependent properties that map the image plane [mm] to pixel coordinates [pixel]:

- o_x, o_y: principal point (image centre) in pixels
- $\alpha = -f'/s_x$, $\beta = -f'/s_y$: magnification factors
- s_x, s_y: pixel size [mm/pixel]
- Aspect ratio $= \alpha/\beta$

Pixel coordinates:

$$u = \alpha \frac{x}{z} + o_x, \qquad v = \beta \frac{y}{z} + o_y$$

In matrix form:

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \frac{1}{z} \begin{pmatrix} \alpha & 0 & o_x \\ 0 & \beta & o_y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

Extrinsic Parameters

Camera location-dependent properties that transform world coordinates to camera coordinates:

- $\mathbf{R}$: 3×3 rotation matrix
- $\mathbf{t}$: 3×1 translation vector

World to camera transformation:

$$\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} \mathbf{R}^T & -\mathbf{R}^T\mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Complete Projection Pipeline

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \frac{1}{z} \underbrace{\begin{pmatrix} \alpha & 0 & o_x \\ 0 & \beta & o_y \\ 0 & 0 & 1 \end{pmatrix}}_{\text{Intrinsic (2D→2D)}} \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\text{Projection (3D→2D)}} \underbrace{\begin{pmatrix} \mathbf{R}^T & -\mathbf{R}^T\mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix}}_{\text{Extrinsic (3D→3D)}} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

Exam Tip: Forward vs Inverse

3D → 2D projection is **well-posed** (unique). 2D → 3D recovery is **ill-posed** (depth lost). Need extra information: stereo vision or scene priors.

2.2.5 Digital Images

Digital Image

A 2D array (matrix) of numbers $I(x, y)$ where each element represents the light intensity at a discrete location called a ***pixel*** (picture element).

Two discretisation processes:

- **Pixelisation**: intensity values averaged within each grid location (spatial sampling)
- **Quantization**: intensity represented using finite discrete values (intensity sampling)

Image Formats:

Format	Values/Pixel	Bits	Range
Binary/Monochrome	1 binary	1	$\{0, 1\}$
Greyscale	1 real	8	$[0, 255]$ or $[0, 1]$
Colour (RGB)	3 reals	24 (8×3)	3 channels × 256 levels

Convention: 0 = black, 255 (or 1.0) = white. Origin at top-left corner.

Format Conversions:

RGB to Greyscale (weighted average):

$$I_{\text{grey}} = \frac{r \cdot I_R + g \cdot I_G + b \cdot I_B}{r + g + b}$$

Greyscale to Binary (thresholding):

$$I_{\text{bin}}(p) = \begin{cases} 1 & \text{if } I_{\text{grey}}(p) > t \\ 0 & \text{otherwise} \end{cases}$$

2.2.6 Camera Sensors and Demosaicing

Charge Coupled Device (CCD)

A semiconductor with a 2D matrix of photo-sensors. Each sensor accumulates charge proportional to incident light intensity and exposure time. Charge pattern corresponds to the image.

Colour from CCDs:

Photo-sensors are not wavelength-selective. Colour is achieved via filters:

Solution	Pros	Cons
3 CCDs + prism	High quality, full sampling	Expensive, bulky
1 CCD + colour mask	Cheap, compact	Coarser sampling, needs demosaicing

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Bayer Mask

The most common colour filter array. Has 2× as many green filters as red/blue (matches human sensitivity). Results in subsampled colour channels that need interpolation.

Demosaicing

The process of filling in missing colour values so each pixel has R, G, and B values. A form of interpolation.

Interpolation Methods:

Method	Description	Trade-off
Nearest Neighbour	Copy adjacent pixel from same channel	Fast, inaccurate
Bilinear	Average of 2 or 4 nearest same-channel pixels	Fast, blurs edges
Smooth Hue Transition	Interpolate ratio (hue) R/G or B/G	Better colour accuracy
Edge-Directed	Interpolate along axis with lowest gradient	Preserves edges

Edge-Directed Interpolation for Green

To calculate G_8:

1. Compute horizontal gradient: $\Delta_H = |G_7 - G_9|$
2. Compute vertical gradient: $\Delta_V = |G_3 - G_{13}|$
3. If $\Delta_H < \Delta_V$: $G_8 = (G_7 + G_9)/2$ (interpolate horizontally)
4. If $\Delta_H > \Delta_V$: $G_8 = (G_3 + G_{13})/2$ (interpolate vertically)
5. If equal: $G_8 = (G_3 + G_7 + G_9 + G_{13})/4$

2.2.7 The Eye

Eye Anatomy:

- **Cornea**: bulk of refraction (fixed focus)
- **Lens**: further refraction, can change shape to adjust focal length
- **Iris**: regulates light amount (like camera aperture)
- **Retina**: contains photoreceptors
- **Optic nerve**: ∼1 million fibres transmit to brain

Photoreceptor Types:

Type	Sensitivity	Colour	Location
Rods	High (dim light)	None (monochrome)	Periphery
Cones (S)	Low (bright light)	Blue (∼420nm)	Fovea
Cones (M)	Low (bright light)	Green (∼540nm)	Fovea
Cones (L)	Low (bright light)	Red (∼570nm)	Fovea

Distribution:

- **Fovea**: no rods, high cone density. #blue $\ll$ #red $\leq$ #green
- **Periphery**: mostly rods, few cones
- **Blind spot**: no photoreceptors (optic nerve exit)

Foveal vs Peripheral Vision:

Property	Fovea	Periphery
Resolution	High (dense receptors)	Low (sparse receptors)
Colour	Yes (cones)	No (rods)
Sensitivity	Low (cones)	High (rods)
Brain resources	Disproportionately large	Smaller

Retinal Processing—Ganglion Cells:

Receptive Field (RF)

The area of visual space from which a neuron receives input. More generally, the stimulus properties that produce a response.

- ∼1 million ganglion cells vs ∼100 million photoreceptors
- Each ganglion cell collects from multiple photoreceptors
- Large convergence in periphery, smaller in fovea (another reason for lower peripheral acuity)

Centre-Surround Receptive Fields

Ganglion cells have centre-surround RFs:

- **On-centre, off-surround**: active if centre brighter than surround
- **Off-centre, on-surround**: active if centre darker than surround

Functional Benefits of Centre-Surround RFs

1. **Edge enhancement**: weak response on uniform regions, strong near edges
2. **Invariance to ambient light**: measures contrast, not absolute intensity
3. **Efficient coding**: only signals where intensity changes ("redundancy reduction")

Centre-Surround Response

On a uniform surface: input to centre and surround cancels ⇒ weak response.

At an edge: input unequal ⇒ strong response (increased or decreased depending on type).

Under different lighting: if illumination doubles, both centre and surround double, so contrast unchanged ⇒ response unchanged.

Colour Opponent Cells:

Ganglion cells combine cone inputs to create colour-opponent responses:

- **Red-Green opponent**: Red ON / Green OFF (or vice versa)
- **Blue-Yellow opponent**: Blue ON / Yellow OFF (yellow = average of R and G cones)

Difference of Gaussians (DoG)

The standard mathematical model for centre-surround RFs:

$$\mathrm{DoG}(x, y) = G_{\sigma_1}(x, y) - G_{\sigma_2}(x, y)$$

where G_σ is a Gaussian with standard deviation σ, and $\sigma_1 < \sigma_2$.

2.3 Worked Examples

Example 1: Colour Constancy Problem

Q: Why is recovering surface colour from an image an ill-posed problem?

A: The luminance $L(\lambda)$ we observe depends on both illuminance $E(\lambda)$ and reflectance $R(\lambda)$: $L = f(E, R)$. We want to recover R (surface colour) but only measure L and don't know E (light source). Different combinations of E and R can produce the same L, making the problem ill-posed.

Example 2: Thin Lens Calculation

Q: A lens has focal length $f = 50$mm. An object is at distance $z = 200$mm. Where does the image form?

A: Using the thin lens equation:

$$\frac{1}{f} = \frac{1}{|z|} + \frac{1}{|z'|} \quad \Rightarrow \quad \frac{1}{50} = \frac{1}{200} + \frac{1}{|z'|}$$

$$\frac{1}{|z'|} = \frac{1}{50} - \frac{1}{200} = \frac{4-1}{200} = \frac{3}{200}$$

$$|z'| = \frac{200}{3} \approx 66.7\text{mm}$$

Example 3: Perspective Projection

Q: A point $P = (30, 40, 100)$ in camera coordinates. Focal length $f' = 10$mm. Find image coordinates.

A:

$$x' = \frac{f'}{z}x = \frac{10}{100} \times 30 = 3\text{mm}$$

$$y' = \frac{f'}{z}y = \frac{10}{100} \times 40 = 4\text{mm}$$

Image point: $P' = (3, 4, 10)$ mm.

Example 4: Intrinsic Parameters

Q: Convert camera coordinates $(x', y') = (3, 4)$mm to pixel coordinates. Given: $s_x = s_y = 0.01$mm/pixel, $f' = 10$mm, image centre $(o_x, o_y) = (320, 240)$.

A:

$$\alpha = -\frac{f'}{s_x} = -\frac{10}{0.01} = -1000, \quad \beta = -\frac{f'}{s_y} = -1000$$

From projection: $x'/z' = 3/10 = 0.3$, $y'/z' = 4/10 = 0.4$

Actually, using the formula directly with x/z:

$$u = \alpha\frac{x}{z} + o_x = -1000 \times 0.3 + 320 = 20$$

$$v = \beta\frac{y}{z} + o_y = -1000 \times 0.4 + 240 = -160$$

(Negative v indicates point projects above image boundary.)

Example 5: Centre-Surround Benefits

Q: Explain why centre-surround RFs provide invariance to ambient lighting.

A: Centre-surround cells measure **contrast**—the difference between centre and surround intensities. If ambient light doubles, both centre and surround receive twice the light, but their *difference* (relative to their sum) remains constant. The response depends on $I_{\text{centre}} - I_{\text{surround}}$, which encodes local contrast, not absolute intensity.

2.4 Summary

Key Takeaways

1. **Image formation**: $L = f(E, R)$—luminance depends on illumination and reflectance; recovering R is ill-posed.
2. **Optics**: Pinhole = sharp but dim; Lens = bright but limited depth of focus. Thin lens: $1/f = 1/|z| + 1/|z'|$.
3. **Projection**: $x' = (f'/z)x$, $y' = (f'/z)y$. Parallel lines $\rightarrow$ vanishing points.
4. **Camera parameters**: Intrinsic (focal length, pixel size, centre) + Extrinsic (rotation, translation).
5. **Digital images**: Pixelisation (spatial) + Quantization (intensity). Bayer mask needs demosaicing.
6. **Eye**: Fovea (high-res, colour, low-sensitivity) vs Periphery (low-res, mono, high-sensitivity).
7. **Ganglion cells**: Centre-surround RFs provide edge enhancement, lighting invariance, efficient coding.

Camera	**Eye**
Uniform pixel grid	Non-uniform (fovea vs periphery)
Bayer mask for colour	3 cone types for colour
Demosaicing interpolates	Brain processes incomplete data
Raw pixel output	Centre-surround preprocessing
Fixed sensitivity	Rods (dim) vs Cones (bright)